

Indomo-Injector: Self-Injection Device for At-Home Acne Treatment



Scarlett Bonner
Nico Del Villar
Emily Burger
Anthony Tran

Kate Huefner (S)
Wilson Zambrano (S)
Caiya Coggs (F)
Daniel Kotkosky (F)

Project Advisor(s): Prof. Sophia Bahena
Project Liaison(s): Rick Bente - CEO
Cara Davis - COO

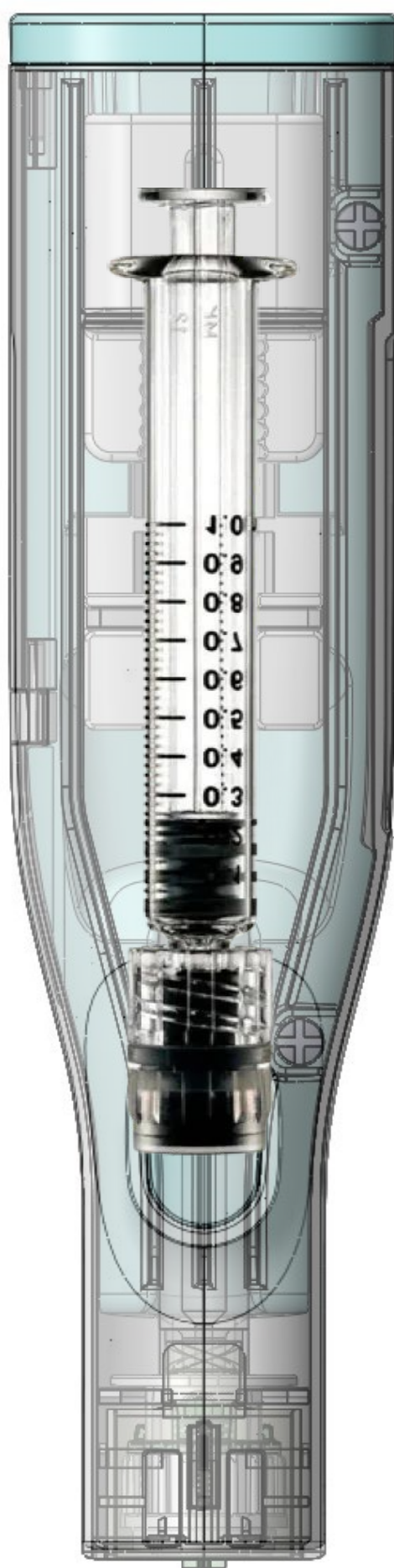
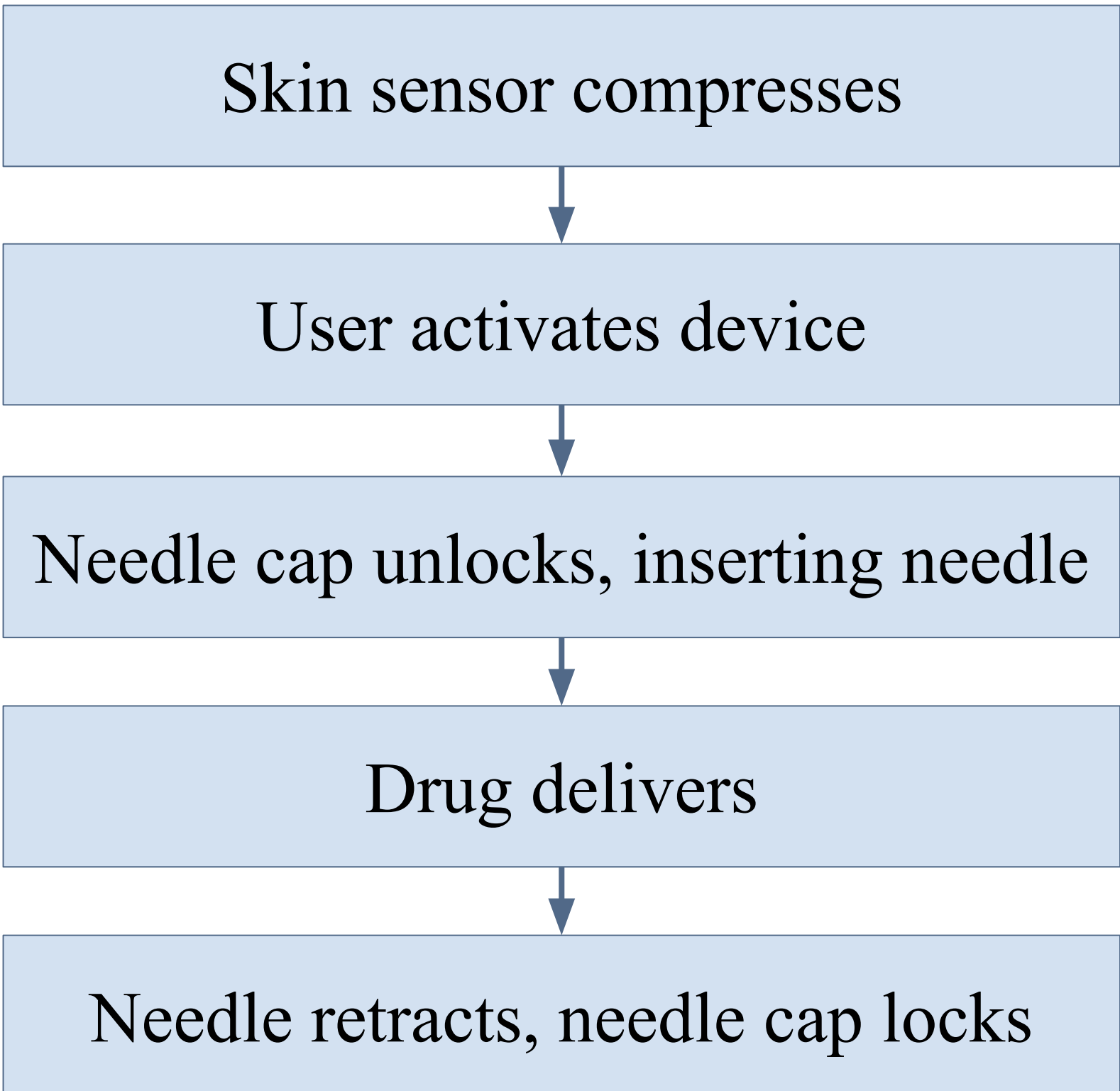


Department of Engineering, Harvey Mudd College, Claremont, CA, USA

Project Overview

Nearly 20 million Americans suffer from **inflammatory acne** each year, but access to effective treatment like corticosteroid injections is **limited by high costs** and **long wait times**. Indomo aims to create a cheaper and **more convenient** solution: a **reusable**, single-dose intradermal injector for **at-home** acne treatment. They tasked the clinic team with **iterating** on their current design.

Process Flow of Injector Use

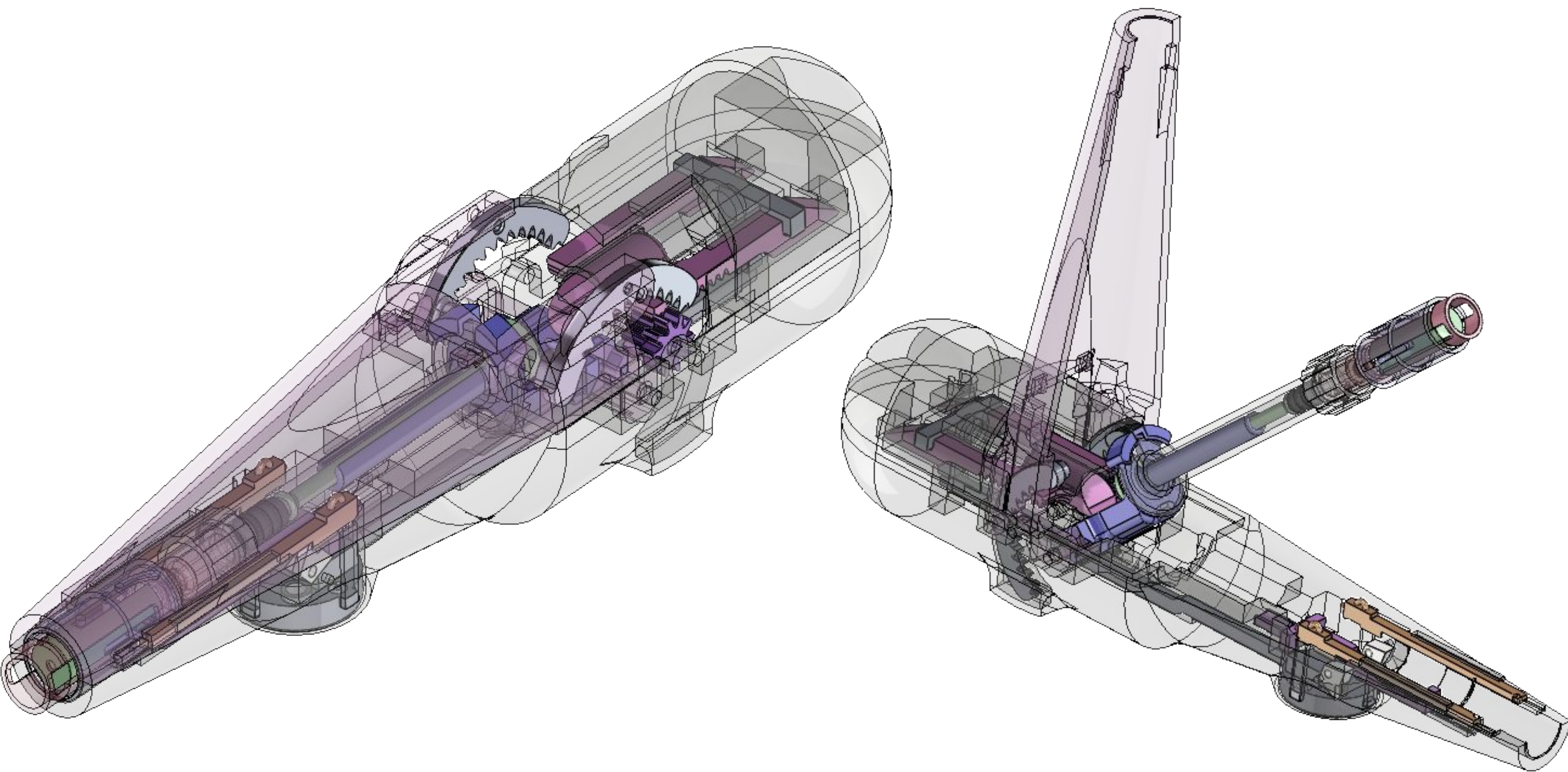


Areas for Improvement

- Objective 1: Reduce nozzle diameter to facilitate ease of aiming
- Objective 2: Accommodate a 1mL Luer lock glass syringe
- Objective 3: Enhance user experience (simplify loading process, reduce noise generation, improve user safety)

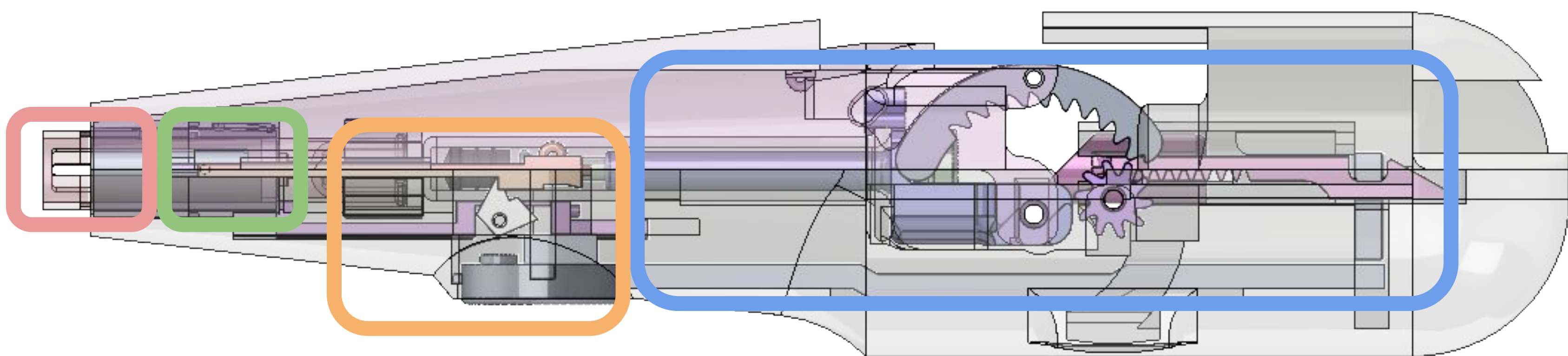


Overall Design

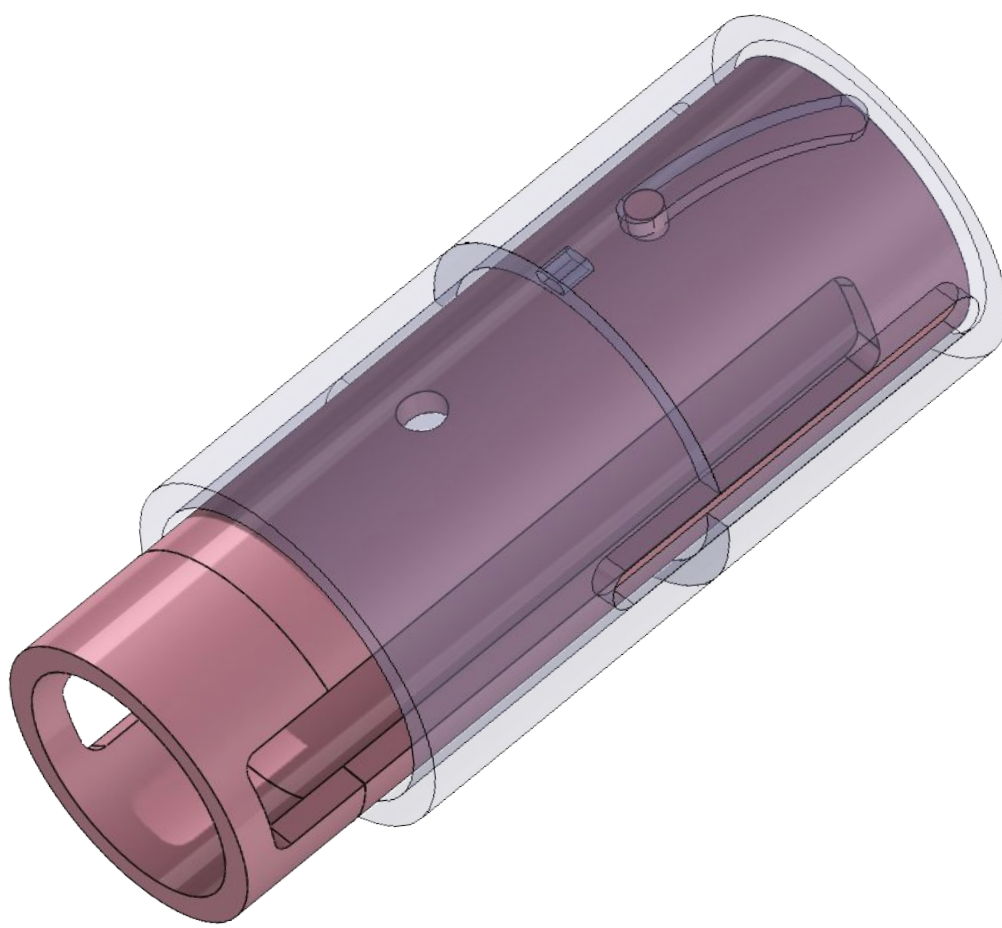


1. Novel needle cap design: simplified mechanical complexity by isolating needle cap movement
2. Reduced nozzle diameter by 6 mm (30% smaller than Indomo's previous design), improving precision
3. Intuitive device loading: resetting is embedded in the syringe loading
4. Selected an underhand grip to improve ergonomic design

Subsystems and Integration

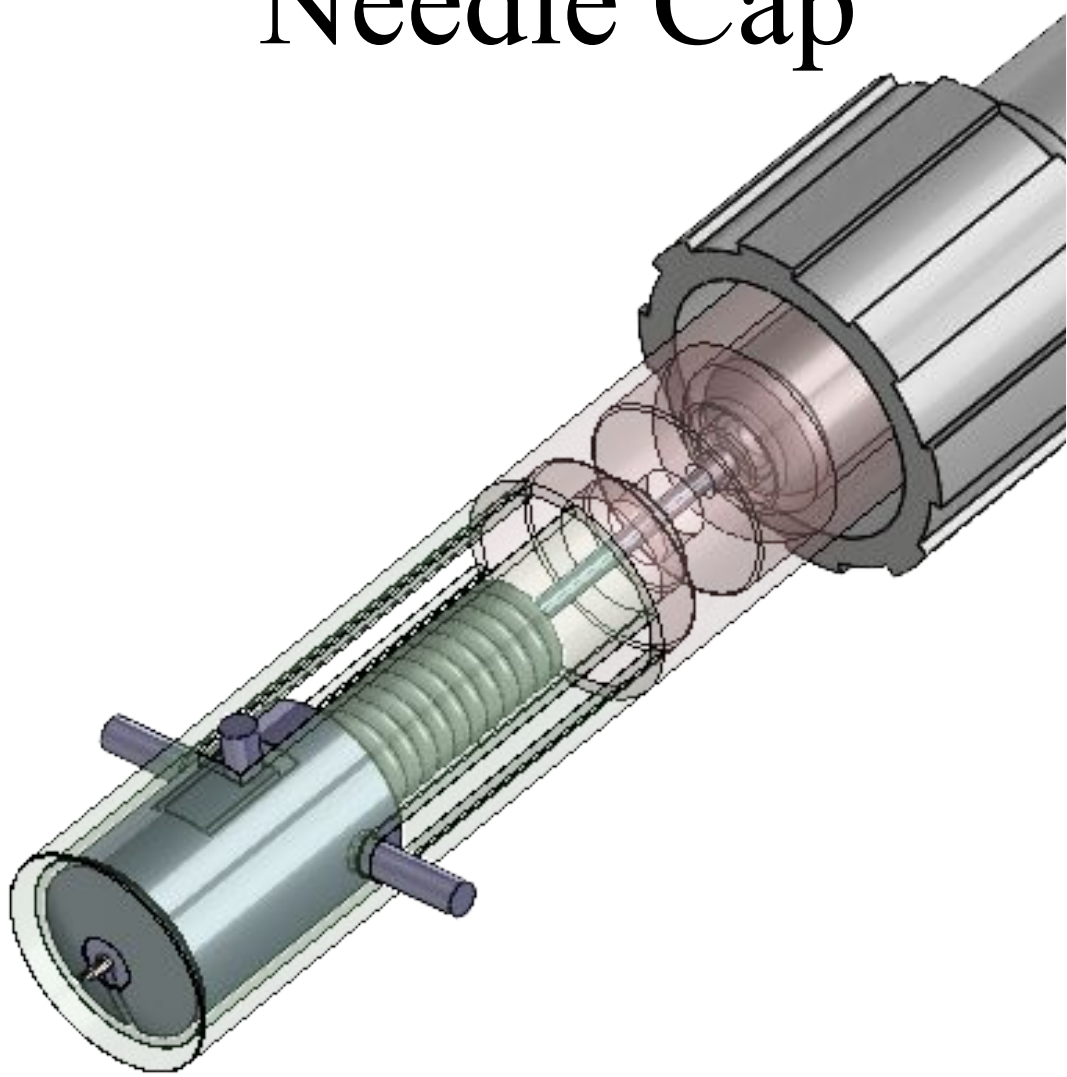


Skin Sensor



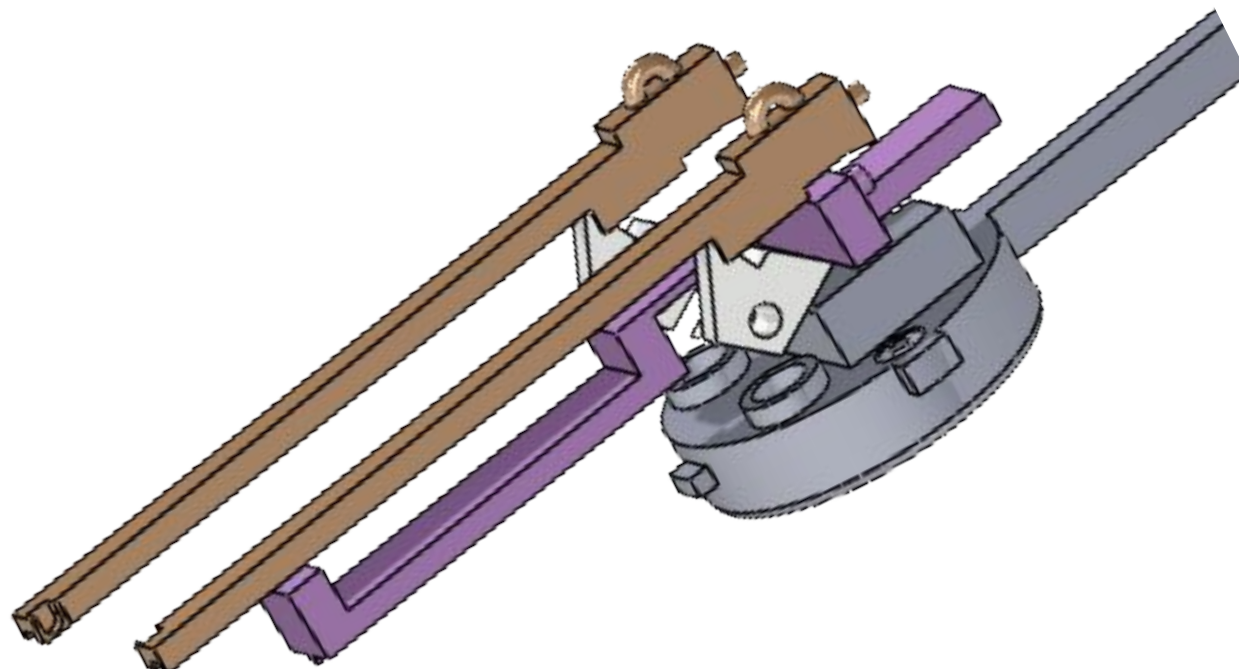
- Smaller radius allows device to press flat against skin with enough force
- Safely locks user activation until device is correctly positioned on skin

Needle Cap



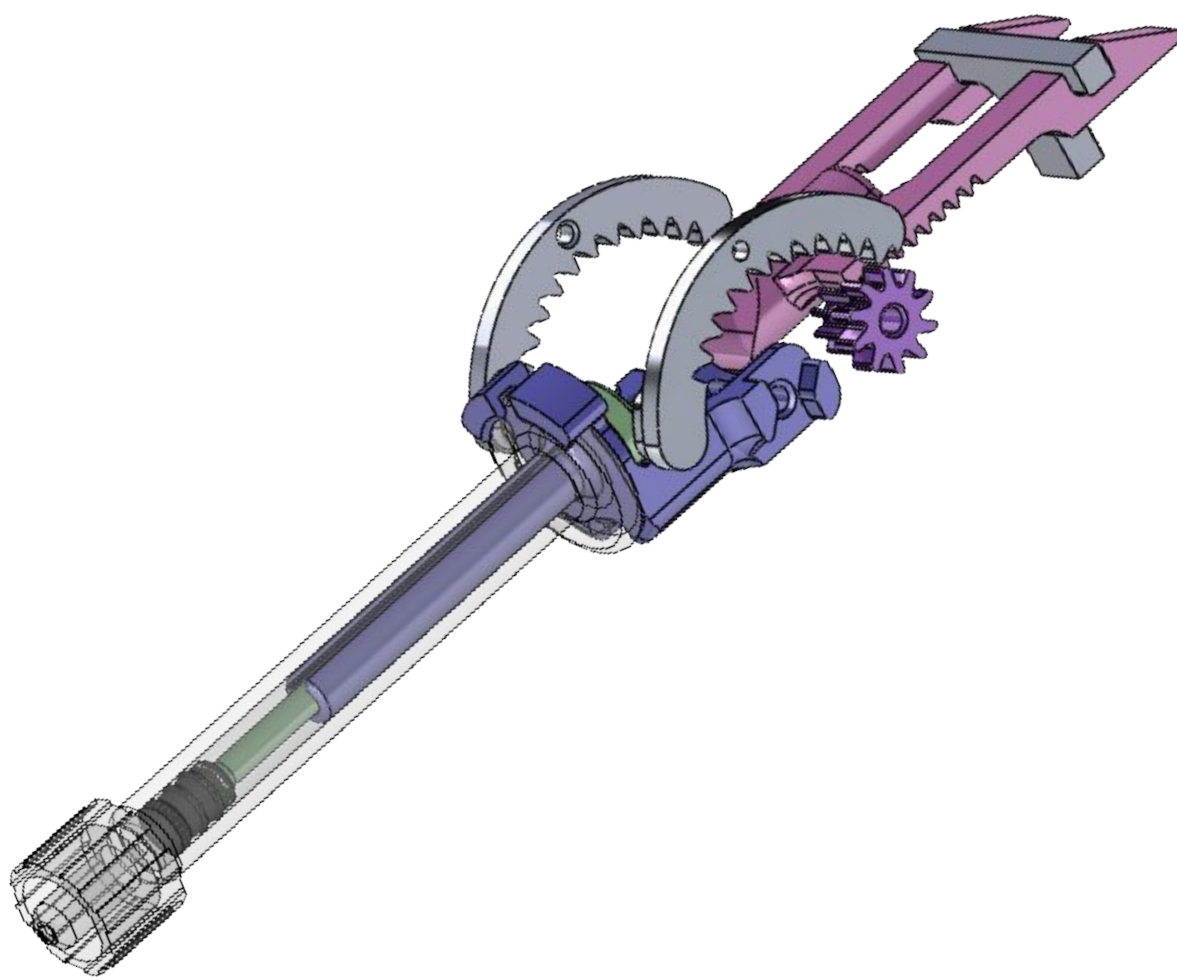
- Protects needle before and after injection by retracting into the device and locking in place
- Exposes and inserts needle into skin during injection

User Activation



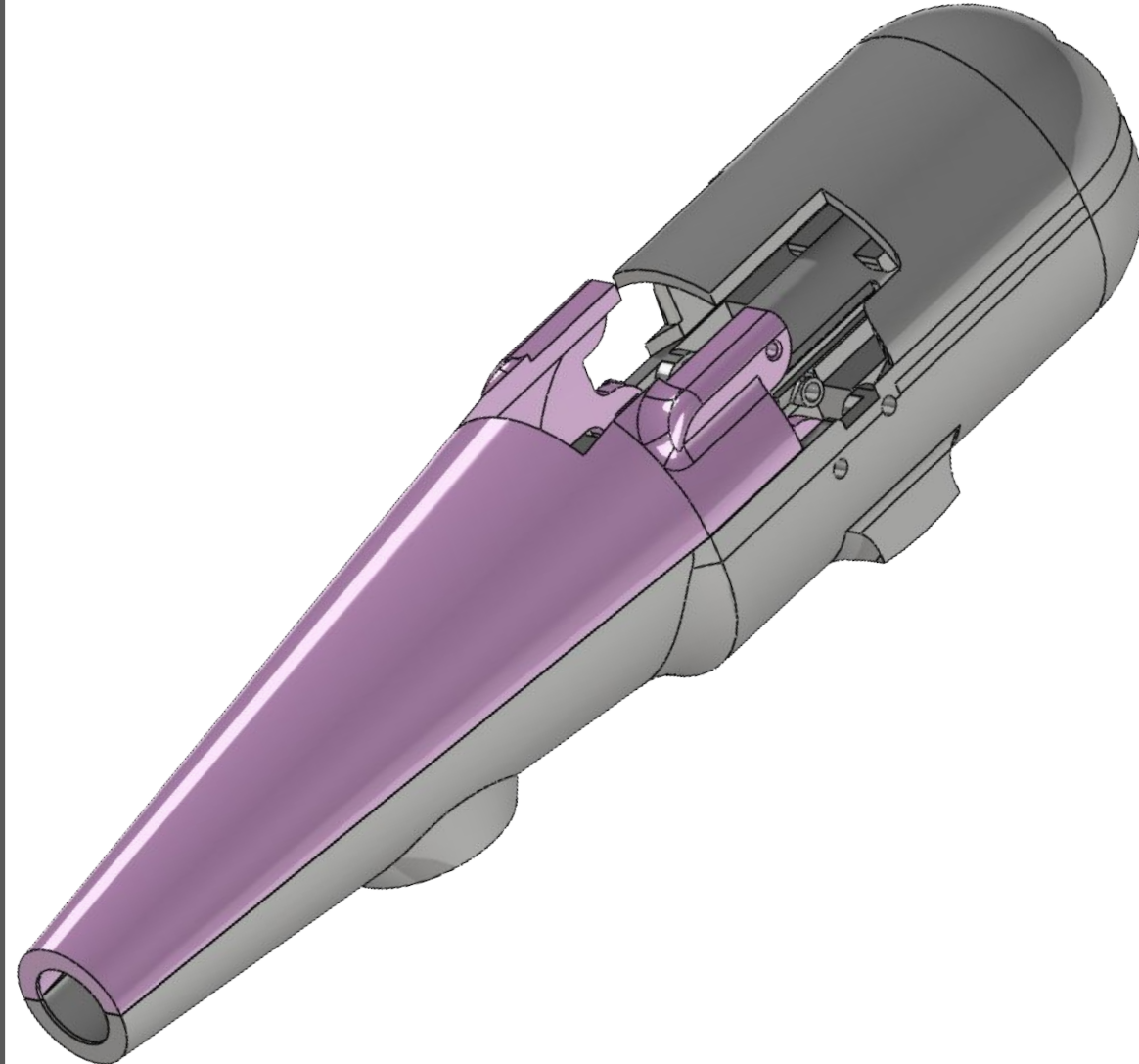
- Interlocks with skin sensor to prevent injection until safe to activate
- Transfers force through yellow arms to activate drug delivery and reveal needle

Drug Delivery



- Holds the designated 1mL prefilled glass syringe
- Intuitive rod to load syringe onto
- Delivers target cc dose of steroid from syringe

Device Housing



- Accommodates all necessary components of device
- Designed ergonomics based on user feedback

Conclusion

The team's design **effectively addresses** the desired changes to the existing device. It introduces a **slimmer concentric needle cap** and skin sensor configuration and **streamlines the user's loading process** by integrating the drug delivery mechanism with the device opening. Additionally, the novel needle cap design **isolates motion** to the cap itself, eliminating the need to move the entire syringe within the device as in previous versions. The final prototype **meets core project specifications** and establishes a **strong foundation for continued development** that will allow the design space to be increased from the previous sprint. Future iterations could further refine the housing to improve both **ergonomics** and the integration of the **hinging components**.

Acknowledgements

Thank you to the Indomo team and the HMC faculty and staff including: Professor Mendelson, Professor Santana, Drew Price, Sydney Torrey, and Jacob Staimpel.